

1. Calibrated EEG

Single channel, 100 Hz
Continuous recording in μV

2. E3 preprocessing

Zero-phase Butterworth filter
Order 4; 0.5–30 Hz
Clip at $\pm 800 \mu\text{V}$; divide by 100

3. Epoch segmentation

30 s per epoch
3,000 samples per epoch

6. Stage classification

Softmax output per epoch
W, N1, N2, N3, REM

5. Non-causal TCN

Six blocks; dilations
1, 2, 4, 8, 16, 32
Receptive field: 253 epochs

4. ResNet-1D encoder

One feature vector per epoch
128 dimensions

